

Lab 3: Architectural Pattern Selection

Hostel Maintenance & Issue Ticketing System

Name : Kuber Vijayakar

SRN : PES1UG24CS671

1. Architectural Choice

We chose Layered Architecture for the Hostel Maintenance & Issue Ticketing System.

The system is organized into three main layers:

Presentation Layer: Provides the Resident and Maintenance Warden portals.

Data Layer: Stores ticket information and encrypted issue photos.

2. Reason 1 – Separation of Concerns

Layered Architecture provides a clear separation of concerns by dividing the system into different layers. Each layer is responsible for a specific set of functions. This makes the system easier to develop, maintain, test, and modify without affecting the other layers.

3. Reason 2 – Centralized Hostel Management

The architecture is suitable for a centralized hostel-management environment. Residents can raise and track maintenance tickets, while the Maintenance Warden can review tickets, assign maintenance staff, and monitor deadlines through a single controlled system. This provides a structured workflow for managing hostel maintenance issues.

4. Security Advantage

A major security advantage is the use of centralized Authentication and Role-Based Access Control (RBAC). Residents can access only tickets associated with their own room, while Wardens and maintenance staff receive permissions based on their assigned roles.

5. Performance Benefit

SLA Monitoring and Escalation can run as background processes. This allows residents to submit tickets and view their ticket status without waiting for SLA checks or notification processing to finish. As a result, the system can provide a more responsive user experience.

6. Conclusion

Therefore, Layered Architecture is an appropriate choice for the Hostel Maintenance & Issue Ticketing System because it provides clear separation of responsibilities, centralized management, improved security through role-based access, and better responsiveness through background SLA processing.